

Initialising the directory

Using Borg with the local repository, repokey encryption and BLAKE2b is often faster, since Borg 1.1 (to know more about why it is faster, refer to the document at <https://blake2.net/blake2.pdf>)

```
$ borg init --encryption=repokey-blake2 /path/to/repo
```

In the above example I used key encryption 'repokey-blake2' to encrypt and the repository path '/path/to/repo' to store the backup files; you can choose any path that has enough space to backup your files.

Here is an example of how to take a backup of the VM:

```
/bin/sh /qnap/backup-scripts/rcm-backup-vm-1st-disk.sh
Windows2012-VM1
```

An output sample is given below:

```
[root@se_vm2 ~]# /backup/backup-scripts/rcm-backup-vm-1st-
disk.sh Window2012-VM1 /home/admin/VM-Storage/vm1-c-100gb.
qcow2
/home/admin/VM-Storage/vm1-d-150gb.qcow2
Domain snapshot tempsnap created
/home/admin/VM-Storage/vm1-d-150gb.tempsnap
Saving backup as Window2012-VM1-2018-07-11
Archive Window2012-VM1-c-2018-07-11 already exists

Block commit: [100 %]
Successfully pivoted
Block commit: [100 %]
Successfully pivoted
Domain snapshot tempsnap deleted
Pruning old backups
Keeping archive: Window2012-VM1-c-2018-07-11 Wed, 2018-07-11
20:43:10
[1068528431bd4a2c 444654f9fba25d51da35
123aee0c71bc062975f76dcf5770]
Done
Target Source
-----
hda /home/admin/VM-Storage/vm1-c-100gb.qcow2
hdb /home/admin/VM-Storage/vm1-d-150gb.qcow2
```

When you see the logs, you will notice that Borg deletes old backed up files after the specified days.

To prune old backups, the code is as follows:

```
borg prune -v --list /backup --prefix "$DOMAIN-" --keep-
within=1m & wait || /bin/true
```

The logs will look something like what follows:

```
Keeping archive: Window2012-VM1-d-2018-07-12 Thu, 2018-07-12
12:40:33
[3601dbf1a489da61 dc05d61d2387c2693 6922e97100c896d0c5f2
42b0fb59352]
Keeping archive: Window2012-VM1-c-2018-07-12 Thu, 2018-07-12
04:00:03
[005c1d93647328f09 37a73363857bd1f151deb942d4c7159a8
d815885620e01f]
Keeping archive: Window2012-VM1-d-2018-07-11 Wed, 2018-07-11
21:40:27
[947c1c62b6163a12 63bab9c6bff1a75ff5b2b86007a521
e1683653e77d285fb3]
Keeping archive: Window2012-VM1-c-2018-07-11 Wed, 2018-07-11
20:43:10
[1068528431bd4a 2c444654f9fba25d51da35123aee0c71bc
062975f76dcf5770]
Done
```

Extracting file from archive

```
# Extract entire archive
```

```
$ borg extract /path/to/repo::my-files
```

```
# Extract entire archive and list files while processing
```

```
$ borg extract --list /path/to/repo::my-files
```

```
# Verify whether an archive could be successfully extracted,
but do not write files to disk
```

```
$ borg extract --dry-run /path/to/repo::my-files
```

```
# Extract the "src" directory
```

```
$ borg extract /path/to/repo::my-files home/USERNAME/src
```

```
# Extract the "src" directory but exclude object files
```

```
$ borg extract /path/to/repo::my-files home/USERNAME/src
--exclude '*.o'
```

```
# Restore a raw device (must not be active/in use/mounted at
that time)
```

```
$ borg extract --stdout /path/to/repo::my-sdx | dd of=/dev/
sdx bs=10M
```



Note: For extraction, you will need a passphrase, as you will be asked for it.

END

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